

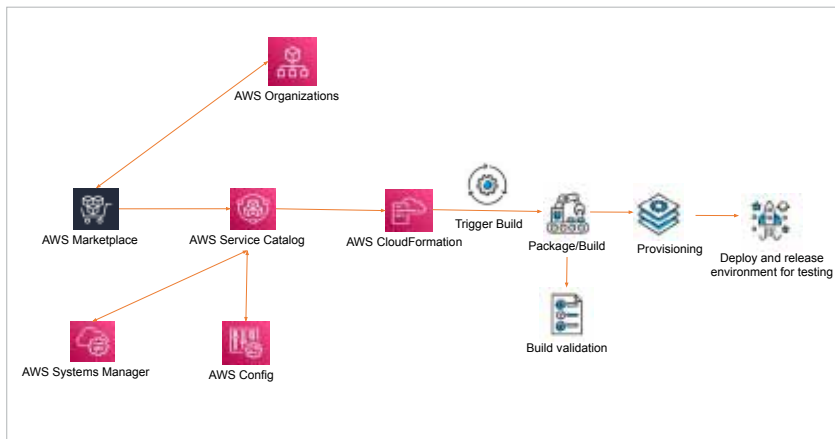


Figure 4: Infra as Code using AWS CloudFormation, Config and Systems Manager

is a comprehensive tool that can be used in combination with Azure Resource Manager (ARM) templates or with Terraform templates and create VHD (virtual hard disk) files of a virtual machine (VM). This can be imported to any cloud platform like AWS EC2 using the VM import tool in AWS or the GCP VM import tool in Google Cloud.

Azure Image Builder has two major functions — to create images from the template builder, and to run the build image with an integrated CI pipeline to deploy it to the target VM in Azure. The image build from this Image Builder service can be a Windows or Linux image, and hence suitable for importing by other cloud service providers as well as for using the VM import tool.

Integrating a CI pipeline with Azure Image Builder is very simple. Integrate it with Azure DevOps or any other customised DevOps pipeline with Ansible, Bamboo or Jenkins. Azure Image Builder also provides role based access control (RBAC) to access the image repository and template repository in a secure way.

AWS CDK has many pre-configured templates available, which can be reused to build a provisioning script. It usually works hand-in-hand

with the AWS CloudFormation service. One has to develop an IaC script using AWS CDK and provisions using CloudFormation.

If you are familiar with Terraform, you can use the `aws-tf` construct library, and if you are familiar with Kubernetes and want to develop a container deployment script, you can use the `aws-cdk8s` construct library and deploy it in the AWS Elastic Kubernetes Engine (EKS).

Nowadays, Low-code/No-code and Infrastructure as Code (IaC) are getting popular as they are less complex and more efficient for operations. The Systems Manager Service of AWS can be used to group cloud resources, get operational insights and automate

operational tasks. It helps to reduce the time and effort used in troubleshooting operational problems. This service can handle orphaned cloud resources and group cloud resources, get monitoring and management activities onto a single UI dashboard quickly, and simplify the operation and management of cloud infrastructure at scale. When you are managing large applications and infrastructure estate in a cloud platform, it is tedious to manage the server provisioning and deployment across different stages. Infrastructure as Code (IaC) simplifies this with template based scripts.

Google Anthos, HashiCorp Terraform and Red Hat CloudForms are leading this space equally for creating, changing and provisioning infrastructure for any cloud platform including service management, security, and compliance and optimisation. When you work with more applications and services during cloud adoption, you may end up with repeated work in infrastructure provisioning, platform setup, and application development and deployment. If you can automate many of these jobs, it will ease the software development journey particularly when you need to repeat activities at different stages like development, testing, pre-production and production. **END** 🐧

References

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